Coursework 1 - Statistical Modelling

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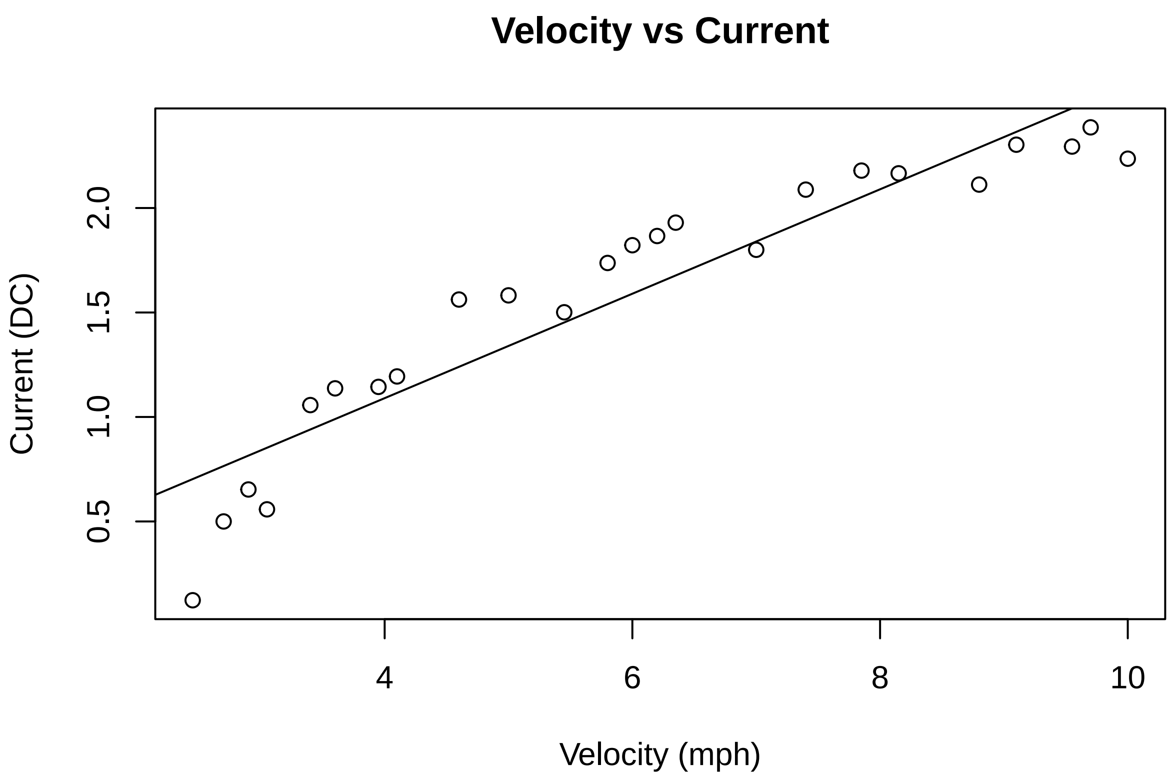
Question 1:

i).

```r
# Reading the data
wind_data <- read.table("windmill.txt", header = TRUE)
head(wind_data)
```

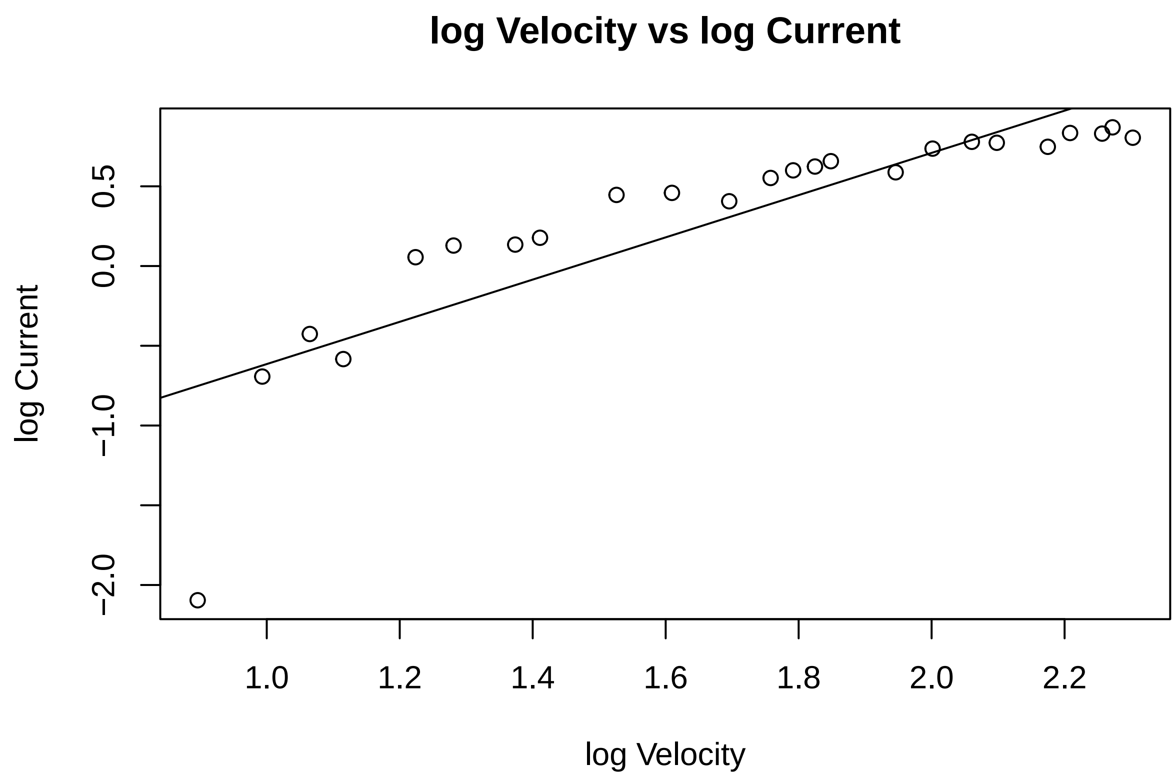
```
##   velocity current
## 1     2.45   0.123
## 2     2.70   0.500
## 3     2.90   0.653
## 4     3.05   0.558
## 5     3.40   1.057
## 6     3.60   1.137
```

```r
# Plot current vs velocity
plot(wind_data$velocity, wind_data$current,
  xlab = "Velocity (mph)",
  ylab = "Current (DC)",
  title("Velocity vs Current"))
abline(lm(current ~ velocity, data = wind_data))
```



From the scatter plot above, we can see that the current increases as velocity increases. However, the points clearly follow a curved pattern rather than a straight line; they lie below the fitted line at low velocities, above it in the middle region, and below it again at higher velocities. This suggests that a linear model would not be a perfect fit and would only ever be an approximation.

```
# Plotting log current vs log velocity
plot(log(wind_data$velocity), log(wind_data$current),
  xlab = "log Velocity",
  ylab = "log Current" ,
  title("log Velocity vs log Current"))
  abline(lm(log(current) ~ log(velocity), data = wind_data))
```



The log plot follows a similar curved pattern to the original plot, and the curvature appears slightly more pronounced. The first data point also deviates noticeably from the rest and may be an outlier. Excluding this first point, the remaining points in the plot appear to lie closer to a straight line pattern than in the original plot. This outlier could make modelling on the log scale more difficult and potentially less accurate.

From visual inspection alone it is difficult to decide which scale gives a better fit for an SLRM. By eye, the original plot appears slightly more linear overall. However, by excluding the potential outlier, the log scale seems to be more linear and therefore the better model to fit a SLRM to. We should use analytical methods to decide which scale is more suitable rather than just visual estimation, as it is hard to judge which scale is better with any degree of certainty.

ii).

```r
# Making model on original scale
fit_original <- lm(current ~ velocity, data = wind_data)

summary(fit_original)
```

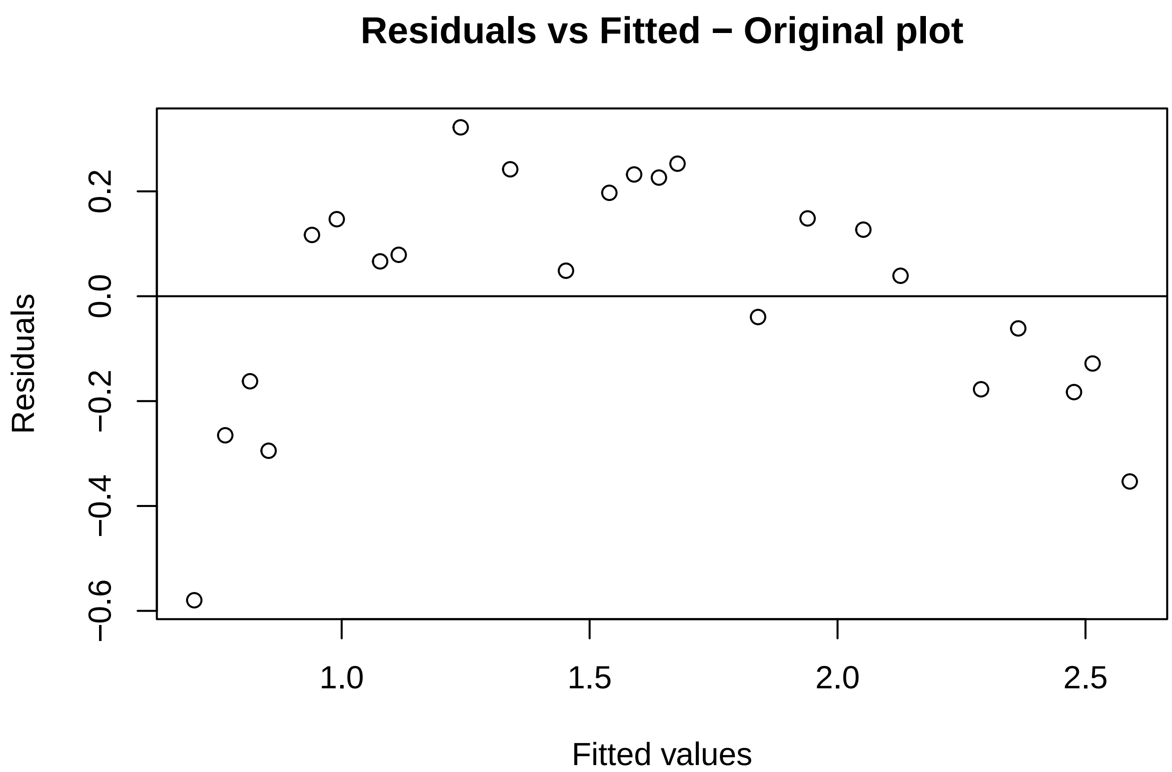
```
##
## Call:
## lm(formula = current ~ velocity, data = wind_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.57974 -0.16600  0.05755  0.16059  0.32203
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.09056    0.12787   0.708    0.486
## velocity     0.24987    0.01991  12.548 1.68e-11 ***
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2325 on 22 degrees of freedom
## Multiple R-squared:  0.8774, Adjusted R-squared:  0.8718
## F-statistic: 157.5 on 1 and 22 DF,  p-value: 1.678e-11
```

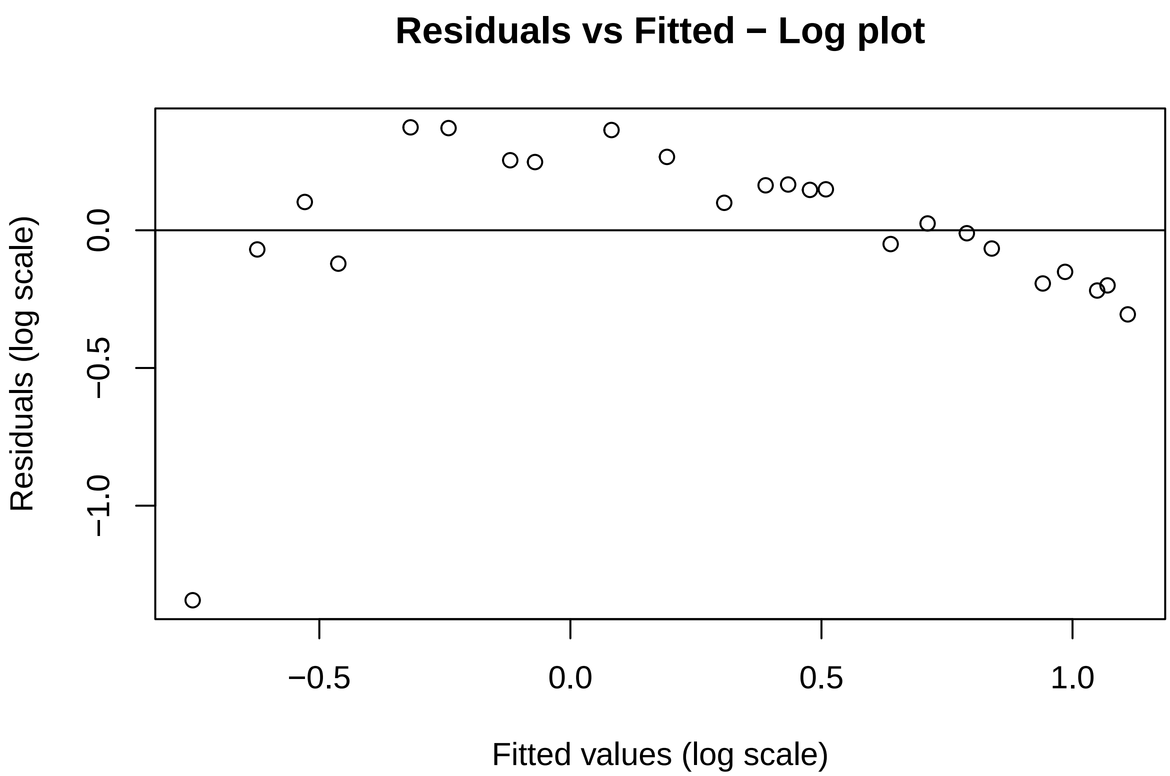
```r
# Making model on log scale
fit_log <- lm(log(current) ~ log(velocity), data = wind_data)

summary(fit_log)
```

```
##
## Call:
## lm(formula = log(current) ~ log(velocity), data = wind_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.34336 -0.12866  0.06227  0.18661  0.37377
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -1.9387     0.2946  -6.580 1.29e-06 ***
## log(velocity)   1.3241     0.1682   7.871 7.73e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3567 on 22 degrees of freedom
## Multiple R-squared:  0.7379, Adjusted R-squared:  0.726
## F-statistic: 61.95 on 1 and 22 DF,  p-value: 7.725e-08
```

```r
# Residual plot 1
plot(fitted(fit_original),
  resid(fit_original),
  xlab = "Fitted values",
  ylab = "Residuals",
  title("Residuals vs Fitted - Original plot"))
  abline(h = 0)
```




```
# Residual plot 2
plot(fitted(fit_log),
  resid(fit_log),
  xlab = "Fitted values (log scale)",
  ylab = "Residuals (log scale)",
  title("Residuals vs Fitted - Log plot"))
  abline(h = 0)
```



From the R^2 values, the original scale model is a better fit ($R^2 = 0.8774$) than the log model ($R^2 = 0.7379$). However, the residual plots for both models show systematic curvature rather than random scatter about the line $y = 0$, indicating that neither model follows the linearity of the mean assumption that we have made. Therefore both SLRMs should be treated as approximations rather than perfect models.

Since both models show evidence of nonlinearity, the original scale model is preferred due to its higher R^2 value.

iii).

```r
# Confidence intervals for model coefficients
summary(fit_original)
```

```
##
## Call:
## lm(formula = current ~ velocity, data = wind_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.57974 -0.16600  0.05755  0.16059  0.32203
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.09056    0.12787   0.708    0.486
## velocity     0.24987    0.01991  12.548 1.68e-11 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2325 on 22 degrees of freedom
## Multiple R-squared:  0.8774, Adjusted R-squared:  0.8718
## F-statistic: 157.5 on 1 and 22 DF,  p-value: 1.678e-11
```

```r
confint(fit_original, level = 0.95)
```

```
##                   2.5 %    97.5 %
## (Intercept) -0.1746181 0.3557356
## velocity     0.2085740 0.2911687
```

```r
# Finding the error variance
s2_hat <- sum(fit_original$res^2) / fit_original$df.residual
s2_hat
```

```
## [1] 0.05405403
```

Error variance and our confidence intervals for model coefficients:

$\hat{\beta}_0$: (-0.1746, 0.3557)

$\hat{\beta}_1$: (0.2086, 0.2912)

$\hat{\sigma}^2$: (0.0541)

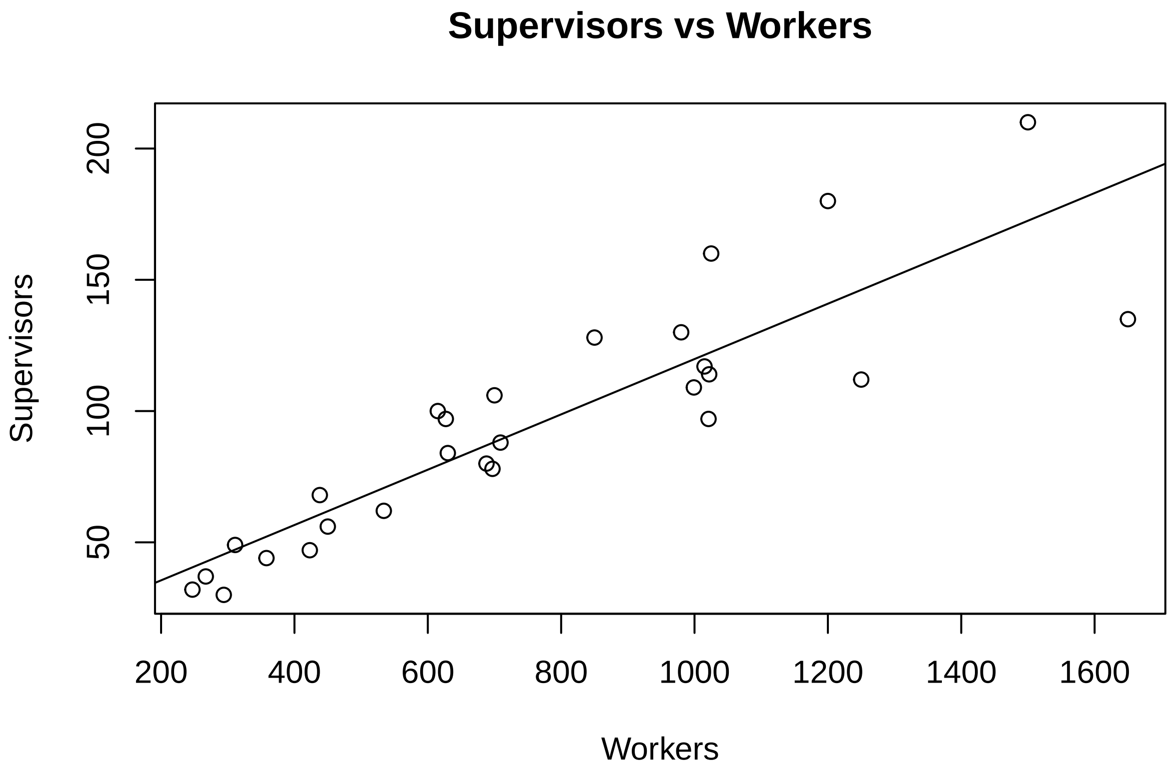
Therefore our final model for the original scale is:

$$\widehat{\text{Current}} = 0.0906 + 0.2499 \cdot \text{Velocity}$$

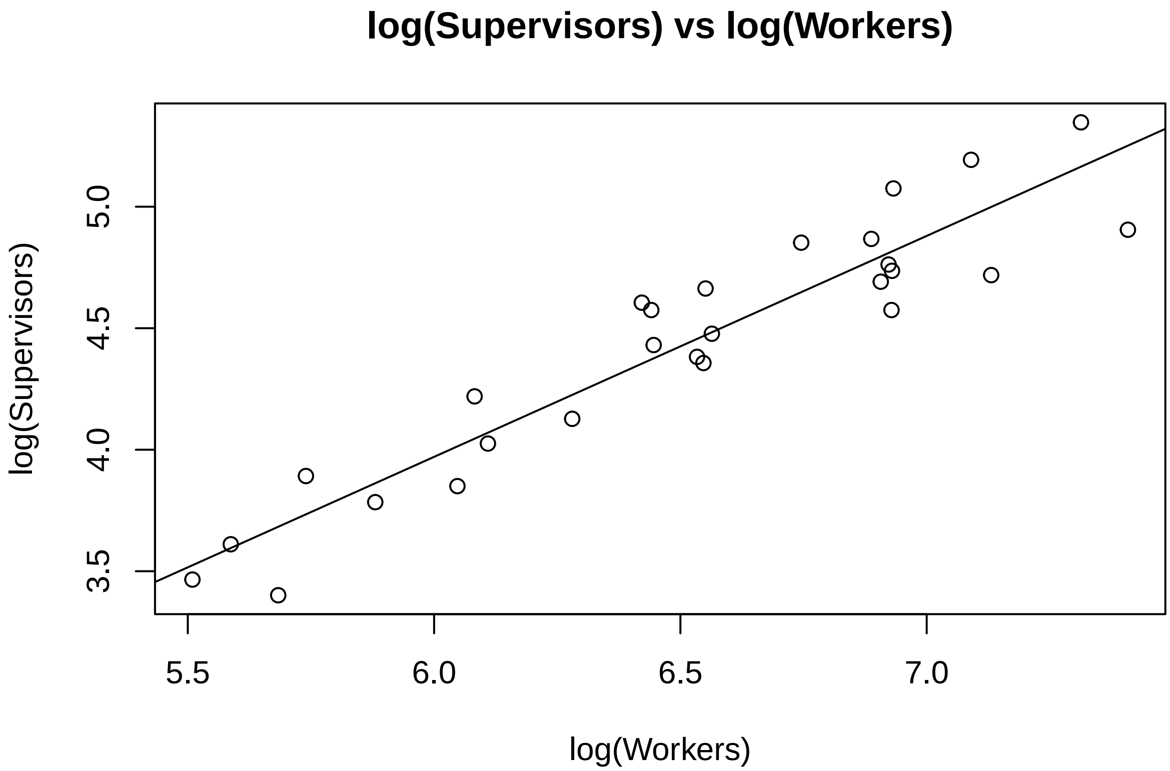
Question 2:

i).

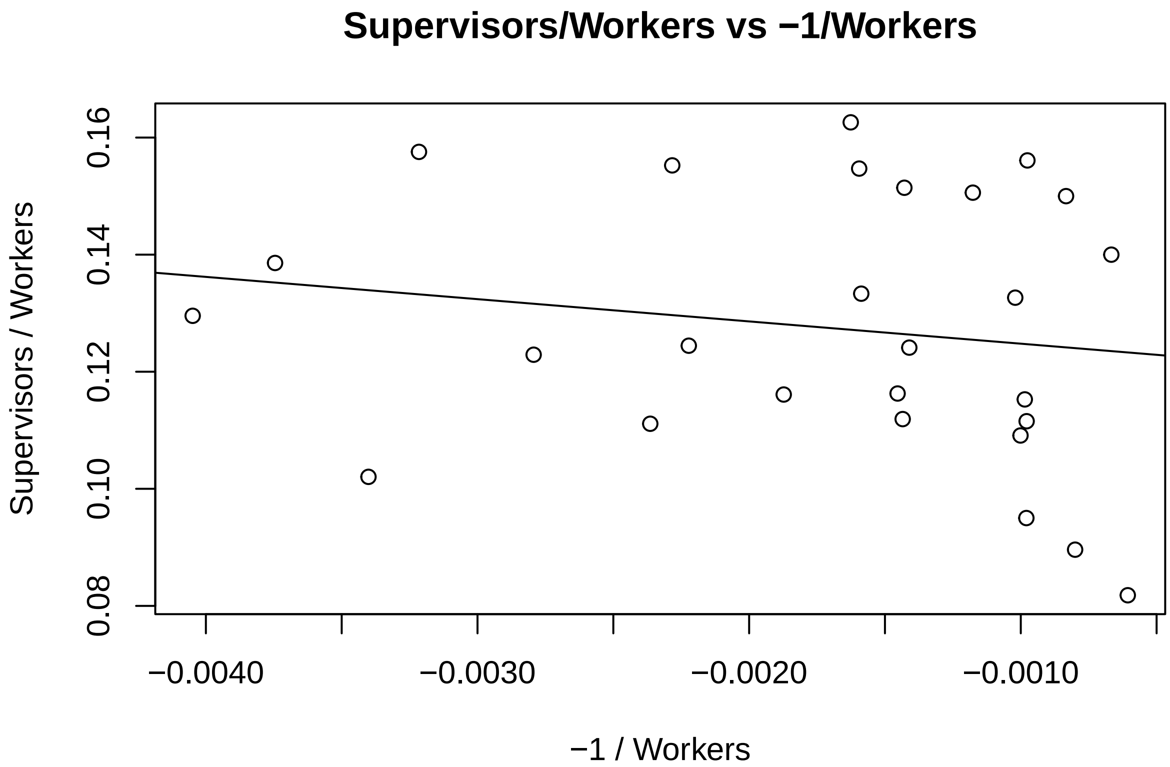
```r
# Reading the data
super <- read.table("super.txt", header = TRUE)

# Scatter plot of Supervisors vs Workers
plot(super$Workers, super$Supervisors,
  xlab = "Workers",
  ylab = "Supervisors",
  title("Supervisors vs Workers"))
  abline(lm(Supervisors ~ Workers, data = super))
```




```r
# Scatter plot of log(Supervisors) vs log(Workers)
plot(log(super$Workers), log(super$Supervisors),
  xlab = "log(Workers)",
  ylab = "log(Supervisors)",
  title("log(Supervisors) vs log(Workers)"))
  abline(lm(log(Supervisors) ~ log(Workers), data = super))
```




```r
# Ratio scatter plot
plot(-1 / super$Workers, super$Supervisors / super$Workers,
  xlab = "-1 / Workers",
  ylab = "Supervisors / Workers",
  title("Supervisors/Workers vs -1/Workers"))
  abline(lm(I(super$Supervisors/super$Workers) ~ I(-1/super$Workers)))
```



On the original scale, the relationship looks very close to being linear. However, it is hard to tell if the trend continues as the number of workers increases due to a 'fanning' of the data points. This fan shape provides evidence of heteroscedasticity which violates the constant variance assumption (variance of error term ϵ_i increases with x_i).

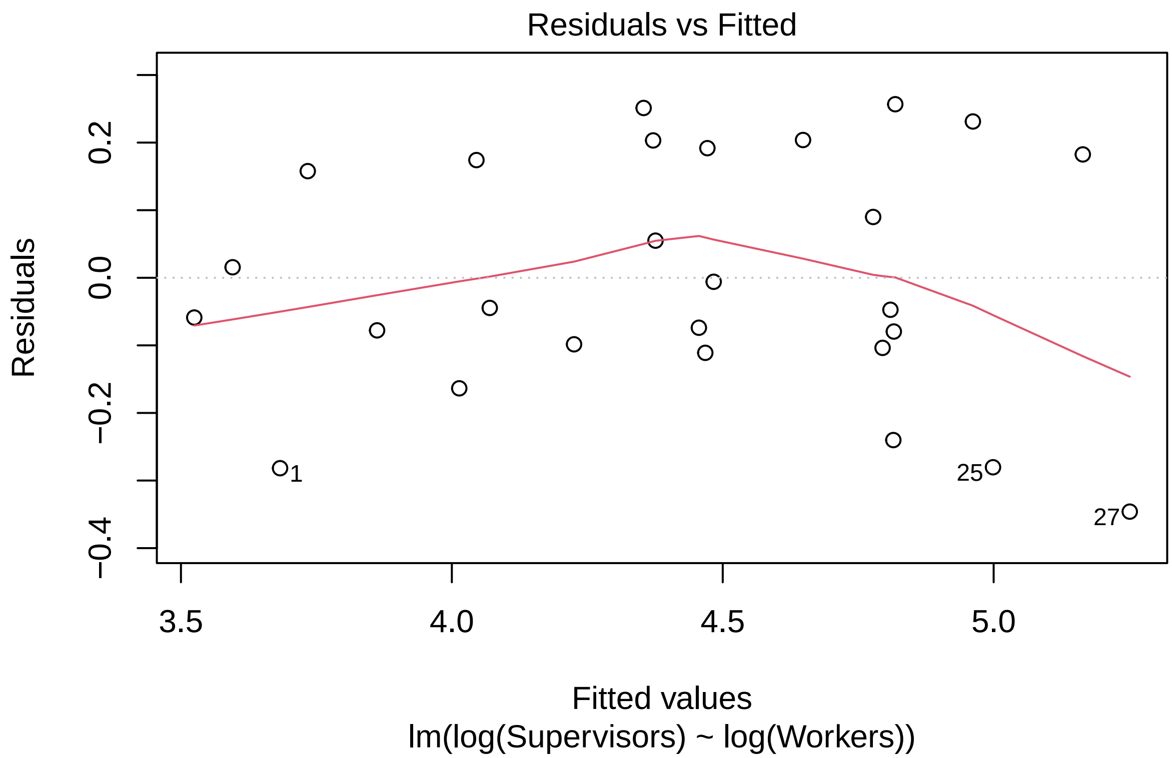
The log plot seems to show a similar linear relationship to the first one and the fanning is severely reduced (if present at all) which suggests that we have nearly constant variance in the error term.

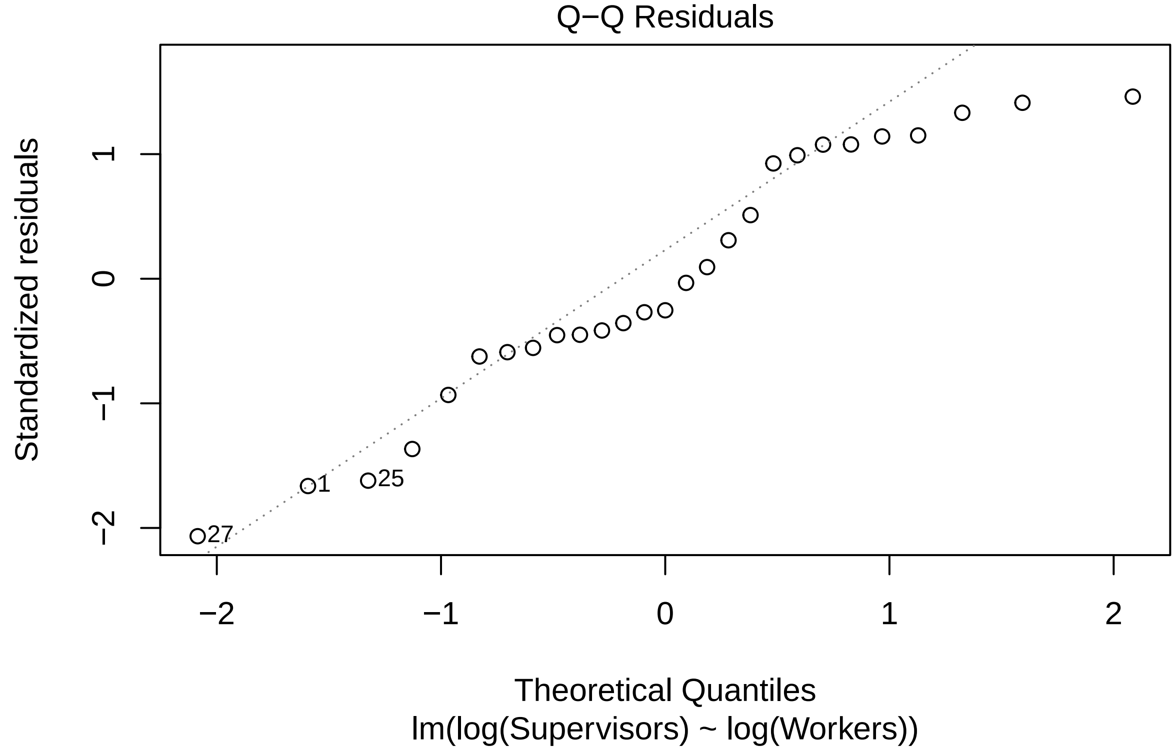
The ratio plot looks to show no clear linear trend, with points looking like they are scattered randomly. Therefore there is little evidence of a linear relationship on this scale and a SLRM would not tell us anything about this plot.

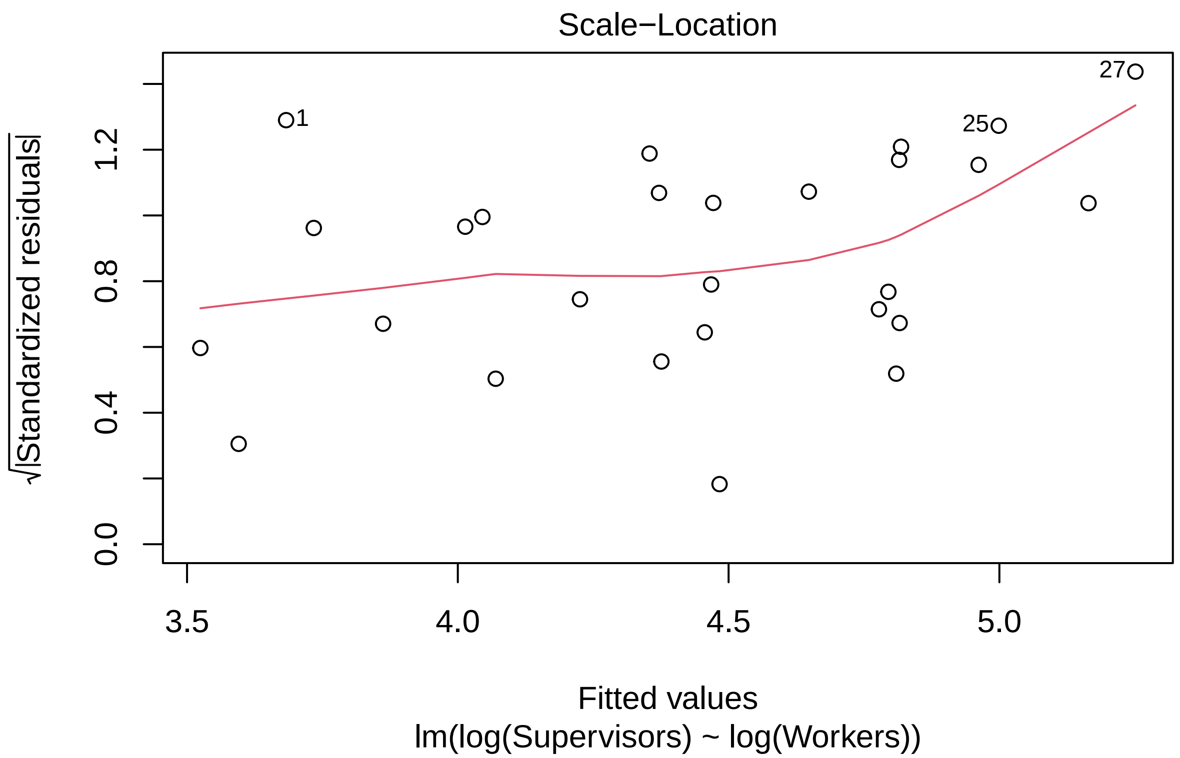
Overall, the log plot is the most suitable to fit a SLRM to, as the relationship appears to be the closest to linear and violates the assumptions of a SLRM the least.

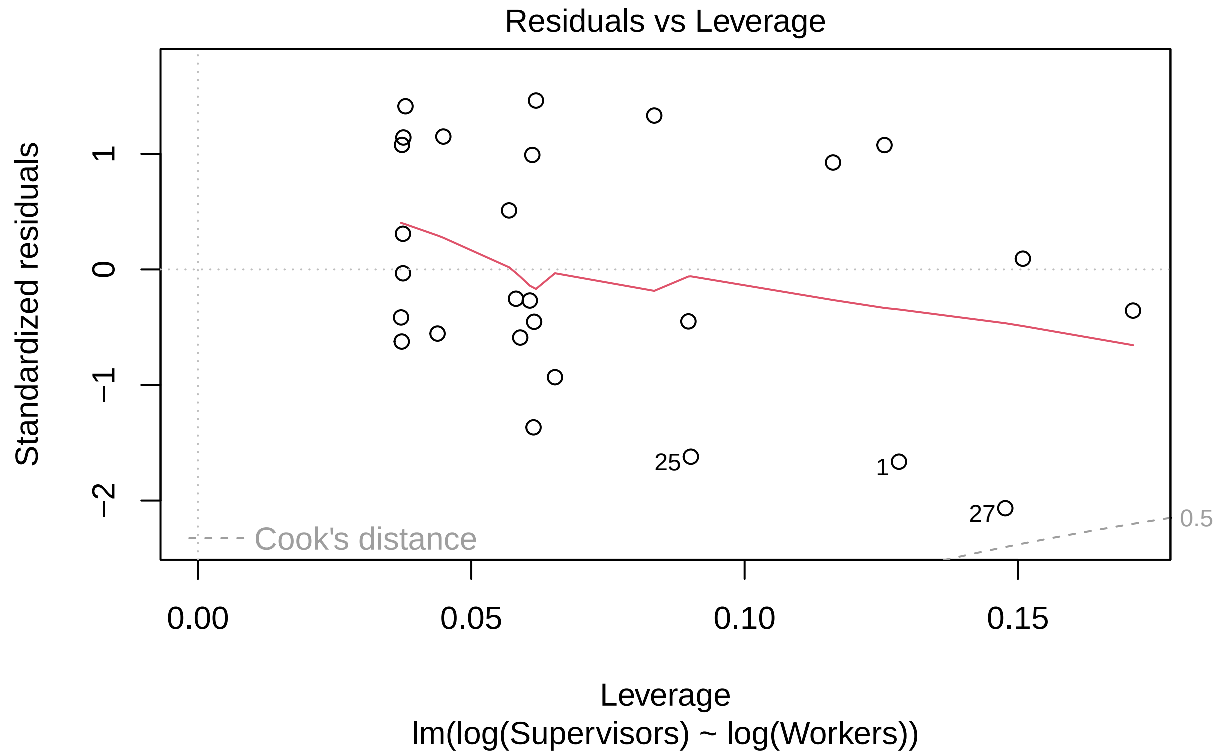
ii).

```
# Fitting a SLRM on the log scale
log_slrm <- lm(log(Supervisors) ~ log(Workers), data = super)
plot(log_slrm)
```


Residuals vs Fitted
Residuals
0.2
0.0
−0.2
−0.4
1
25
27
3.5
4.0
4.5
5.0
Fitted values
lm(log(Supervisors) ~ log(Workers))


Q−Q Residuals
Standardized residuals
1
0
−1
−2
27
1
25
−2
−1
0
1
2
Theoretical Quantiles
lm(log(Supervisors) ~ log(Workers))






```
summary(log_slrm)
```

```
##
## Call:
## lm(formula = log(Supervisors) ~ log(Workers), data = super)
##
```

```
## Residuals:
##     Min      1Q  Median      3Q     Max
## -0.3460 -0.1011 -0.0446  0.1783  0.2568
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -1.48458    0.43544  -3.409  0.00221 **
## log(Workers)  0.90920    0.06673  13.625 4.51e-13 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1814 on 25 degrees of freedom
## Multiple R-squared:  0.8813, Adjusted R-squared:  0.8766
## F-statistic: 185.6 on 1 and 25 DF,  p-value: 4.508e-13
```

Our SLRM will be in the form of:

$$\log(\widehat{\text{Supervisors}}) = \hat{\beta}_0 + \hat{\beta}_1 \log(\text{Workers}) + \varepsilon_i, \qquad \varepsilon_i \overset{\text{ind}}{\sim} N(0, \sigma^2).$$

Where $\hat{\beta}_0$ and $\hat{\beta}_1$ are found from the code above:

$$\hat{\beta}_0 = -1.4846 \qquad , \qquad \hat{\beta}_1 = 0.9092$$

Giving us our model of:

$$log(\widehat{\text{Supervisors}}) = -1.4846 + 0.9092 \log(\text{Workers})$$

By looking at the plots, we can see that 3 extreme points have been labelled as potential outliers, the Residuals vs Leverage graph shows that none of these points exceed within a Cook's distance of 0.5 and are therefore not classed as influential.

When testing the fit of this model, we performed a hypothesis test with the following hypotheses: $H_0 : \beta_1 = 0$ and $H_1 : \beta_1 \neq 0$. The test gives a p-value of 4.51×10^{-13} which is far below the 5% significance level we used. We therefore reject H_0, indicating that there is evidence of a relationship between the variables ($\beta_1 \neq 0$). The diagnostics we ran also give an R^2 value of 0.8813 which suggests that the model explains a large proportion of variation seen in the plot and could be a good fit.

Looking at the Residuals vs Fitted plot, the residuals are fairly well scattered about the line $y = 0$, although there is a slight curvature. Since the magnitude of this curvature is small, the SLRM provides a reasonable approximation of the data.

Finally, by looking at the Q-Q plot, we can see that the trend is very close to linear. However, there seems to be some deviation on the higher end of the plot; suggesting that the normality assumption may not be perfectly met. Overall the Q-Q plot still provides evidence of a well fitted model as any possible deviations are small.

Since most of the SLRM assumptions are well met and has a high R^2 value, it is therefore a good fit for the observed data.

iii).

```r
# Applying log function to data
super$logWorkers <- log(super$Workers)
super$logSupervisors <- log(super$Supervisors)
super$logWorkers_2 <- super$logWorkers^2
```

```r
# Applying quadratic regression model
log_qrm <- lm(logSupervisors ~ logWorkers + logWorkers_2, data = super)
summary(log_qrm)
```

```
##
## Call:
## lm(formula = logSupervisors ~ logWorkers + logWorkers_2, data = super)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.24593 -0.13481 -0.05418  0.14821  0.27512
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -9.0967     5.0246  -1.810   0.0828 .
## logWorkers     3.2927     1.5690   2.099   0.0466 *
## logWorkers_2  -0.1853     0.1219  -1.520   0.1415
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1768 on 24 degrees of freedom
## Multiple R-squared:  0.8917, Adjusted R-squared:  0.8827
## F-statistic: 98.85 on 2 and 24 DF,  p-value: 2.592e-12
```

```r
# Comparing variances between QRM and SLRM
log_slrm <- lm(logSupervisors ~ logWorkers, data = super)
anova(log_slrm, log_qrm)
```

```
## Analysis of Variance Table
##
## Model 1: logSupervisors ~ logWorkers
## Model 2: logSupervisors ~ logWorkers + logWorkers_2
##   Res.Df     RSS Df Sum of Sq      F Pr(>F)
## 1     25 0.82242
## 2     24 0.75016  1  0.072254 2.3116 0.1415
```

Using an ANOVA comparison test with $H_0 : \beta_2 = 0$ and $H_1 : \beta_2 \neq 0$ (where β_2 is the coefficient of the $(log(Workers))^2$ term), we obtained $F = 2.3116$ and $p = 0.1415$. Since $p = 0.1415 > 0.0500$, we do not reject H_0 as there is insufficient evidence that the quadratic term improves the model. The F- statistic is not large enough for us to reject H_0 and is therefore not statistically significant at the 5% level.

We can also see that the adjusted R^2 value of 0.8827 is slightly higher than that of the SLRM (adjusted $R^2 = 0.8766$).

This change in the adjusted R^2 value is small so we should base our decision on the hypothesis result which tells us that the QRM does not improve the model and so the SLRM is preferred.

iv).

```r
summary(log_slrm)
```

```
##
## Call:
## lm(formula = logSupervisors ~ logWorkers, data = super)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
```

```
## -0.3460 -0.1011 -0.0446  0.1783  0.2568
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -1.48458    0.43544  -3.409  0.00221 **
## logWorkers   0.90920    0.06673  13.625 4.51e-13 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1814 on 25 degrees of freedom
## Multiple R-squared:  0.8813, Adjusted R-squared:  0.8766
## F-statistic: 185.6 on 1 and 25 DF,  p-value: 4.508e-13
```

```r
# 95% CI for the slope beta_1 calculations
beta_1_hat <- coef(log_slrm)[2]
se_beta_1  <- summary(log_slrm)$coefficients[2, 2]

#Table of parameter estimates
confint(log_slrm, level =0.95)
```

```
##                  2.5 %     97.5 %
## (Intercept) -2.3813973 -0.5877672
## logWorkers   0.7717665  1.0466352
```

```r
# 95% CI results
beta_1_hat - qt(0.975, df = 25) * se_beta_1
```

```
## logWorkers
##  0.7717665
```

```r
beta_1_hat + qt(0.975, df = 25) * se_beta_1
```

```
## logWorkers
##   1.046635
```

Our fitted SLRM is given by:

$$\widehat{\log(\text{Supervisors})} = -1.4846 + 0.9092 \log(\text{Workers}).$$

To test whether there is a relationship between the variables we have $H_0 : \beta_1 = 0$ and $H_1 : \beta_1 \neq 0$. The p-value is 4.508×10^{-13}, so we reject H_0 at the 5% level ($4.508 \times 10^{-13} < 0.0500$). Therefore, there is strong evidence of a relationship between log(Workers) and log(Supervisors).

We also obtain a 95% confidence interval for the slope which is (0.7718, 1.0466) as well as the estimate of the error variance which is $\hat{\sigma}^2 = 0.0329$ with 25 degrees of freedom.

v).

```r
# Change data to log scale
log_super <- data.frame(logWorkers = log(1600))

# 95% CI for the mean of log(Supervisors)
a <- predict(log_slrm,
             newdata = log_super,
             interval = "confidence",
             level = 0.95)
a
```

```
##        fit      lwr      upr
## 1 5.223283 5.083372 5.363193
```

```r
# Converting back to original scale
exp(a)
```

```
##        fit      lwr      upr
## 1 185.5422 161.3171 213.4052
```

```r
# 95% prediction interval for log(Supervisors)
b <- predict(log_slrm,
             newdata = log_super,
             interval = "prediction",
             level = 0.95)
b
```

```
##        fit      lwr      upr
## 1 5.223283 4.824393 5.622172
```

```r
# Converting back to original scale
exp(b)
```

```
##        fit      lwr      upr
## 1 185.5422 124.5109 276.4892
```

For the chosen log model:

$$\log(\widehat{\text{Supervisors}}) = \hat{\beta}_0 + \hat{\beta}_1 \log(\text{Workers}) + \varepsilon_i, \qquad \varepsilon_i \overset{\text{ind}}{\sim} N(0, \sigma^2).$$

The fitted relationship on the original scale is:

$$\widehat{\text{Supervisors}} = 0.2266 \ \cdot \text{Workers}^{0.9092}.$$

Confidence interval for estimated mean of log(Supervisors): (5.0834, 5.3632)

Confidence interval for estimated mean of Supervisors : (161.3171, 213.4052)

If we had 1600 workers, the predicted mean number of supervisors is $\hat{S} = 185.5422$, with a 95% confidence interval of (161.3171, 213.4052).

Supervisors Prediction interval: (124.5109, 276.4892)

We need to find a 95% prediction interval instead of a confidence interval as we are judging whether a single observation of the number of supervisors would be surprising instead of a mean number of supervisors (larger uncertainty in a single observation compared with a mean estimate). From the code we get a prediction interval of (124.5109, 276.4892). Therefore, 256 supervisors would not be surprising as it lies within this interval.